

LUCKNOW, India

WMO: 423690

Lat: 26.761N

Lon: 80.889E

Elev: 410

StdP: 14.48

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.6	46.2	36.0	31.5	71.0	39.0	35.5	67.3	14.4	67.7	12.6	67.9	1.4	270	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.1	109.1	74.2	106.1	74.2	103.8	74.7	85.7	93.9	84.7	92.9	84.0	91.9	8.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
84.3	182.5	89.5	82.8	173.6	88.5	82.4	171.2	88.2	51.0	94.0	49.8	92.2	48.8	92.3	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.4	12.5	10.8		WB	39.3	112.4	2.6	2.7	37.5	114.3	36.0	115.9	34.5	117.4	32.7	119.4
				38.6	87.8	2.1	2.3	37.1	89.4	35.9	90.7	34.7	92.0	33.2	93.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.2	58.5	66.0	75.6	86.2	91.1	91.2	86.4	85.8	84.7	80.0	70.9	61.9
	DBStd	11.72	4.92	4.67	5.30	4.57	4.17	5.02	3.33	2.65	3.01	3.99	3.68	4.61
	HDD50	6	5	0	0	0	0	0	0	0	0	0	0	1
	HDD65	360	205	38	1	0	0	0	0	0	0	0	2	114
	CDD50	10313	269	449	794	1087	1273	1237	1130	1109	1041	929	626	369
	CDD65	5193	5	67	330	637	808	787	665	644	591	464	178	17
	CDH74	69103	69	694	3823	9255	12664	12141	8770	8222	7097	4708	1472	188
CDH80	38491	3	171	1941	5902	8446	7911	4426	3875	3206	2176	426	8	
Wind	WSAvg	4.0	3.3	4.0	4.5	5.1	5.4	5.5	4.9	4.4	4.0	2.3	2.1	2.4
Precipitation	PrecAvg	40.9	0.6	0.5	0.3	0.2	0.8	4.2	12.0	12.5	7.5	1.7	0.2	0.4
	PrecMax	84.2	2.6	2.2	1.7	1.4	2.8	13.2	30.0	25.7	22.4	11.3	1.7	3.1
	PrecMin	19.4	0.0	0.0	0.0	0.0	0.0	0.0	4.2	1.8	0.6	0.0	0.0	0.0
	PrecStd	16.5	0.7	0.6	0.4	0.4	0.8	3.6	6.9	5.9	5.7	3.0	0.4	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.2	89.7	100.8	109.5	112.8	112.8	102.4	98.7	98.7	97.2	89.6	80.9
		MCWB	62.8	69.8	69.5	69.8	74.9	77.1	82.9	85.2	82.3	78.5	69.2	63.4
	2%	DB	75.5	84.4	96.8	106.1	109.3	109.1	98.4	95.3	95.4	94.8	86.4	78.6
		MCWB	60.5	66.2	69.6	70.0	73.9	76.6	83.1	83.9	81.5	76.7	68.2	62.0
	5%	DB	73.0	81.2	93.3	103.9	106.4	105.8	95.3	93.5	93.5	91.8	84.5	76.2
		MCWB	58.8	64.1	68.5	70.3	74.1	77.4	83.0	83.3	81.2	75.5	67.2	61.3
	10%	DB	69.9	78.5	90.0	100.7	104.0	102.4	93.3	91.7	91.7	89.8	82.5	73.5
		MCWB	57.9	62.6	67.2	70.3	74.3	78.7	82.9	82.8	80.7	74.6	66.0	60.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.6	71.1	74.2	80.0	85.0	86.3	87.0	87.4	85.5	83.3	73.6	67.4
		MCDB	75.7	83.2	93.4	97.6	100.1	99.3	94.7	94.2	93.2	91.9	83.0	76.1
	2%	WB	63.0	68.3	72.0	77.4	82.9	84.6	85.5	85.6	84.2	80.8	71.3	65.0
		MCDB	71.6	81.1	91.0	95.4	98.6	97.0	92.7	92.1	91.8	88.7	81.1	73.8
	5%	WB	61.0	66.2	70.2	75.5	80.9	83.6	84.5	84.6	83.1	79.2	69.8	63.2
		MCDB	68.8	78.1	88.5	93.4	96.1	95.2	91.6	91.0	89.9	86.7	80.3	71.9
	10%	WB	59.3	64.2	68.7	73.7	79.4	82.6	83.7	83.6	82.2	77.8	68.3	61.6
		MCDB	66.8	74.7	86.2	92.9	95.1	93.1	90.3	89.6	88.3	85.3	78.4	70.3
Mean Daily Temperature Range	5% DB	MDBR	20.8	23.9	27.3	28.2	24.1	18.1	11.9	11.5	13.3	20.4	24.6	22.7
		MCDBR	26.1	26.9	30.1	31.3	28.1	24.0	15.4	14.3	16.5	21.5	25.4	25.8
	5% WB	MCWBR	12.4	11.1	10.3	10.3	9.3	7.5	6.1	6.1	6.2	8.0	10.1	11.9
		MCDWR	21.6	24.6	28.2	25.7	22.5	17.6	13.2	13.0	13.9	16.9	22.4	22.4
Clear-Sky Solar Irradiance		taub	0.565	0.533	0.520	0.598	0.695	0.744	0.674	0.615	0.590	0.662	0.674	0.638
		taud	1.705	1.828	1.894	1.758	1.610	1.561	1.710	1.861	1.886	1.653	1.571	1.593
		Ebn at Noon	209	234	248	233	211	200	213	226	226	197	179	181
		Edh at Noon	66	63	62	73	84	88	76	65	62	74	75	71
All-Sky Solar Radiation	RadAvg	955	1399	1847	2094	2121	1815	1449	1489	1510	1462	1171	947	
	RadStd	144	115	62	77	102	184	123	118	133	84	69	85	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page